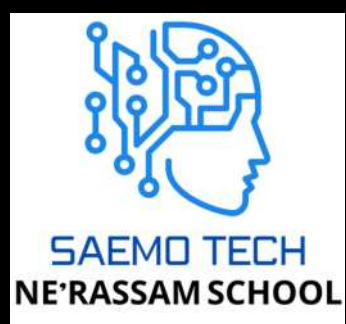




A Survey on Camera-Based Measurement Systems: Techniques, Applications, and Challenges

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Introduction

problem statement:

Identifying key challenges limiting the adoption of camera-based measurement systems across diverse fields.

Objective:

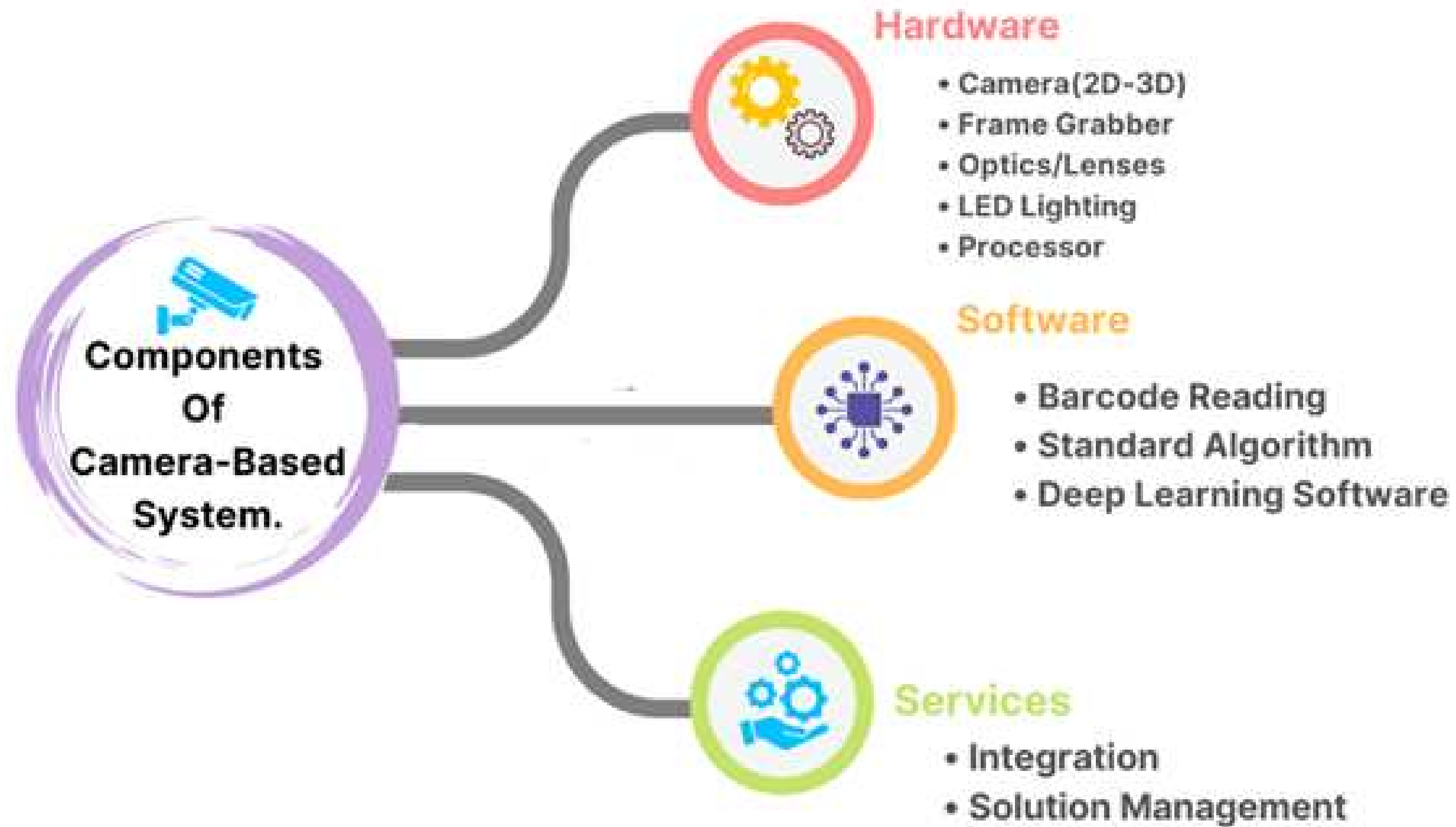
To provide a comprehensive review of camera-based measurement systems

Contributions:

1. The paper provides a systematic classification of camera-based measurement structures in terms of:
 - Sensing techniques.
 - Processing models.
 - Configurations by application.
2. A critical analysis was conducted through more than 30 studies to compare different technical approaches in terms of:
 - accuracy, speed, and robustness.
3. Identification of three main innovation tracks:
 - Adaptive calibration.
 - Integration of hybrid sensors.
 - Advanced AI-based applications.



Components of Camera-Based Systems



Principles:



PROCESSING

Image Processing:

- **Edge detection – Structural measurement.**
- **Thresholding – Liquid level measurement.**
- **Morphological operations – Image enhancement.**



Computer Vision:

- CNNs and Transformers.
- Over 95% accuracy in industrial and medical applications.

STEREO VISION

Stereo Vision:

- Micron-level depth measurement using stereo cameras.
- Sensitive to calibration and temporal synchronization.



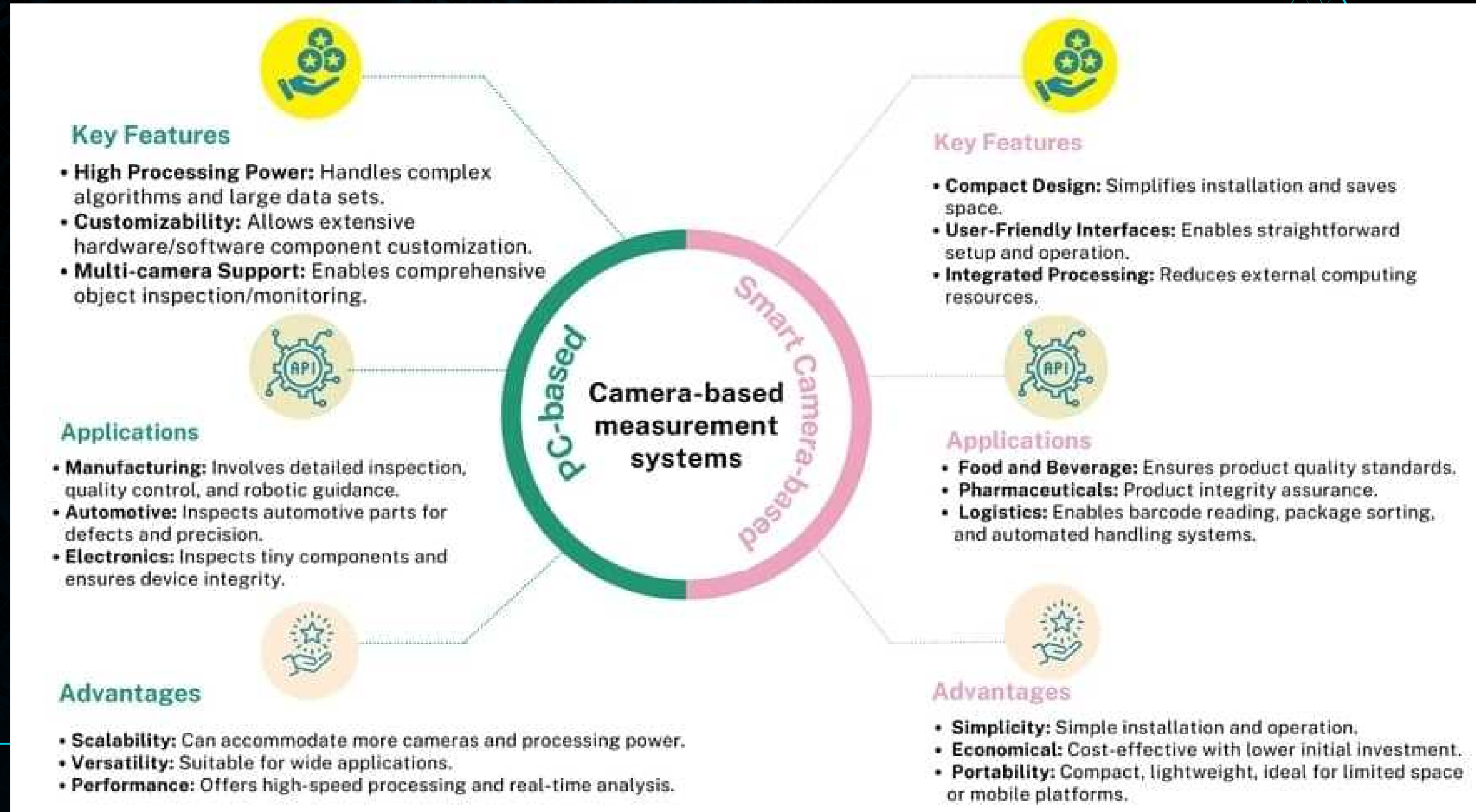
Supporting Factors:

- Lighting.
- Optical accuracy.
- Calibration – Essential for mapping pixels to real dimensions.

Classification of Camera-Based Systems



Architecture-Based Classification



Classification of Camera-Based Systems



Methodological Classification

Traditional Techniques



VS

Deep Learning



Classification of Camera-Based Systems

✓ Applications-Based Classification



Environmental Monitoring.



Structural Health Monitoring.



Industrial Automation.



Healthcare Applications.



Safety and Social Surveillance.

Results

PERFORMANCE COMPARISON ACROSS METHODOLOGICAL APPROACHES

Technique	Accuracy	Processing Rate	Hardware Requirements	Environmental Robustness
Otsu Thresh- olding	±2 cm	60 fps	CPU-only	Low
Stereo Vision	98.19%	20.6 fps	FPGA/ARM	Medium
YOLOv8	98.86%	27 fps	GPU	High

DOMAIN-SPECIFIC OPTIMIZATION STRATEGIES

Application	Primary Metric	Typical Accuracy	Processing Speed	Preferred Tech- nique
Structural Monitoring	Precision	40-50 µm	5-10 min	Photogrammetry
Safety Sys- tems	Frame Rate	98%	27 fps	YOLOv8
Fluid Mea- surement	Cost	±2 cm	60 fps	Thresholding
Environmental	Coverage	0.33 m	3 min/ha	UAV SfM

Conclusion

Challenges:

CHALLENGES

Environment



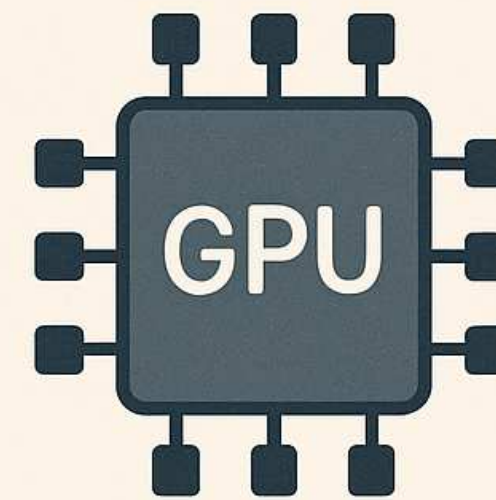
Lighting/noise
reduce accuracy
by 15–25%

Calibration



Takes 15–30 minutes,
prone to error

Computational load



Deep models
require GPU and
large memory

Conclusion

Emerging Solutions:

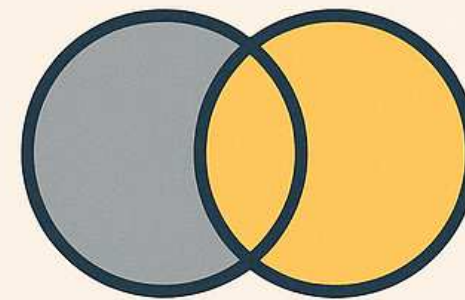
EMERGING SOLUTIONS

Self-calibration



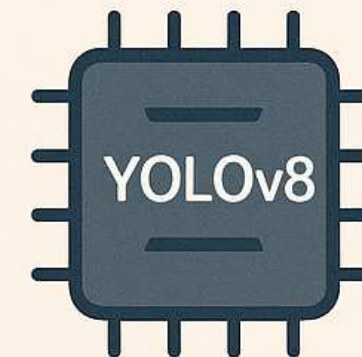
Reduce time to under two minutes.

Multispectral fusion



Increases stability in low-light by 22%

Advanced processors



Implement YOLOv8 with power consumption under 10 W.

FUTURE DIRECTIONS



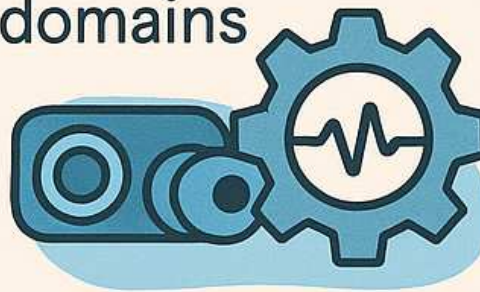
Self-calibrating systems to reduce time (under 2 minutes)



Few-shot learning techniques for fast generalization across domains



Development of unified calibration standards to enhance system

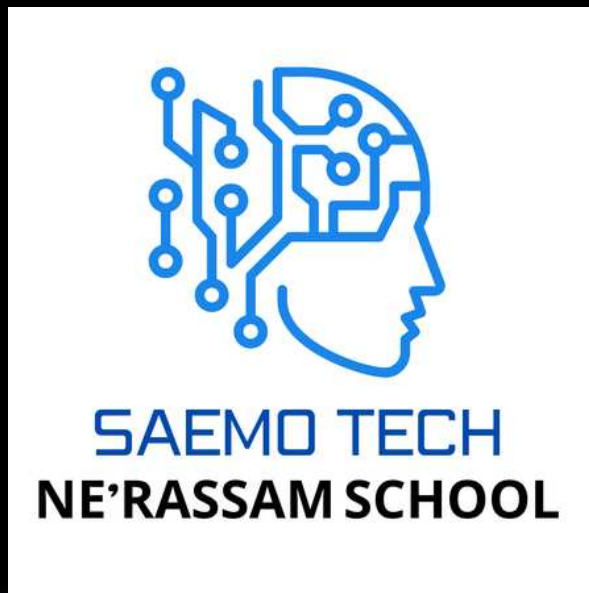


Development of unified calibration standards to enhance system

Conclusion

Future Directions:

Thank You



KEYS SUPPORTING FACTORS

